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REVIEW PAPER

Effective Interventions in Teaching Employment Skills to Individuals with Developmental Disabilities: A Single-Case Meta-analysis

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Abstract

This meta-analysis analyzed 39 studies that met the inclusion criteria for employment-related interventions for individuals with developmental disabilities. Each experiment included in this analysis either met or met with reservations all of the basic design standards (Kratochwill et al. 2010, 2013). Tau-*U* effect sizes were calculated for each A–B contrast extracted from the included experiments, and moderator analyses were conducted for dependent variables, participant characteristics, setting characteristics, and implementer characteristics for all video modeling interventions. Moderate to strong effects were seen across almost all moderator levels, and few significant differences were determined between the levels. According to overall effect sizes, all interventions were effective, but video modeling interventions were considered not only to be effective but also evidence-based interventions in teaching employment skills to individuals with developmental disabilities.

Keywords Meta-analysis · Employment · ASD · ID · Developmental disabilities

Many individuals diagnosed with autism spectrum disorder (ASD), intellectual disabilities (ID), and/or other developmental disabilities (DD) have significant deficits in employment skills crucial for job acquisition and maintenance (Grigal and Deschamps 2012; Hendricks 2010; Hendricks and Wehman 2009). These deficits can create obstacles for many individuals with DD when transitioning from school to work and typically result in unemployment or noncompetitive employment after leaving high school (Carter et al. 2012; Hanley-Maxwell and Izzo 2012). In addition, these individuals rarely receive the necessary transition and employment skill services to address these deficits (Dolyniuk et al. 2002; Hendricks and Wehman 2009). Although regulations and resources are in place to support transition into employment, people with DD continue to attain and retain employment at a lower rate than their peers (Migliore and Butterworth 2008; Wehmeyer et al. 2006). According to successive reports on employment by The National Longitudinal Transition Study-2, 30 to 49% of individuals with disabilities such as ASD, ID, and multiple disabilities were currently employed at 4, 6, and 8 years after leaving high school (Newman et al. 2009, 2011; Sanford et al. 2011). This lack of employment for individuals with disabilities can lead to living dependently upon others (i.e., living with parents, guardians, or caretaker), an inability to earn a stable income, and a decrease in or very little community involvement (Billstedt et al. 2005; Hughes 2001).

In the past 10 years, special education transition practices have been developed to include goals, curricula, and interventions focused on teaching employment skills to promote future employment acquisition and maintenance for students with disabilities. The Individuals with Disabilities Education Improvement Act (IDEA 2004) and No Child Left Behind Act (NCLB 2001) require the use of systematic instruction to teach the skills necessary to prepare students for future employment, beginning no later than when the child turns 16 years old. Employment skills, also referred to as vocational skills (e.g., product assembly, repetitive office tasks, cleaning, participating in an interview), fall into a broader category of "life skills," "functional living skills," or "transition skills" (Alwell and Cobb 2009; Ninci et al. 2015; Palmer 2014).

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While definitions of what is meant by "employment skills" differ, they generally pertain to the accuracy, performance, and fluency of tasks required in the workplace (e.g., completing work tasks correctly and independently, interacting with customers or coworkers, solving problems, and transitioning between work tasks; Palmer 2014; Secretaries Commission on Achieving Necessary Skills [SCANS] 1991; Wehmeyer et al. 2006). A previous meta-analysis that analyzed specific employment skill interventions suggests that employment-related skills include workplace social skills (i.e., asking for help, solving problems, helping a customer), transition between work tasks, independent or accurate job task completion, cooking or preparing food as a work task, assembling a product, and cleaning as a work task (Wehmeyer et al. 2006).

Due to the low incidence of many developmental disabilities, single-case experimental design (SCED) has been applied as it allows for an analysis of outcomes on an individual level (Gast and Ledford 2014; Horner et al. 2005; Kratochwill et al. 2013; Ninci et al. 2015; Palmen et al. 2012; Roth et al. 2014; Walker et al. 2010). Individualized analyses of participants and behaviors provide more information regarding the impact of participant characteristics, target behavior components, overall social significance, and intervention components (Horner et al. 2005). Specific interventions, included in the previous employment intervention categories, include (a) video modeling, (b) audio cueing, (c) visual supports, and (d) systematic prompting. These employment skill interventions have only been analyzed individually through SCED (e.g., Gilson et al. 2017; Hedley et al. 2017), and there remains a need to compare effectiveness across multiple interventions to assess the efficacy of specific intervention techniques in relation to each other (Lipsey and Wilson 2001).

Similarly, definition of the dependent variable can influence outcomes. Previous meta-analyses have focused on broad definitions of functional living skills (i.e., employment skills, independent living skills, social skills, challenging behavior, and self-help skills; Ninci et al. 2015; Roth et al. 2014). Each skill set that falls into the functional living skills definition contains many levels, which makes it difficult to analyze the effects of the intervention related to the complexity or unique behaviors associated with the skill. Participant characteristics also affect the functional relation between the independent and dependent variables (Council for Exceptional Children [CEC] 2014; Gast and Ledford 2014; Horner et al. 2005). Participant characteristics that are most commonly considered are diagnosis, age, and functioning level (CEC 2014; Horner et al. 2005). Previous employment intervention reviews and meta-analyses have solely focused on individuals with a diagnosis of ASD and/or individuals with a functioning IQ of 70 or above (Ninci et al. 2015; Palmen et al. 2012; Roth et al. 2014; Taylor et al. 2012). These participant characteristics are very narrow and hinder the application of these interventions to a broader population of individuals with disabilities. Finally, the context of a SCED study (including the location or setting, the materials used during the procedures, and the individuals carrying out the procedures) can influence intervention outcomes. There is a need to evaluate moderators of intervention effectiveness across independent variables, dependent variables, demographic variables, and contextual variables.

Effect size measures in SCED are crucial to the systematic analysis of the functional relation between independent and dependent variables (Kratochwill et al. 2013). Nonoverlap effect size measures are becoming more common when analyzing comparisons of data and are more accurate for nonparametric analyses of SCED studies compared to parametric measures (e.g., regression; Parker et al. 2011b). Parametric measures follow certain assumptions (e.g., normal distribution, constant variance) that are not typically present in SCED results, and outliers that are prevalent in SCED heavily impact parametric measures (Parker and Vannest 2009; Parker et al. 2011b). To avoid errors that would occur when using parametric analyses in SCED, previous meta-analyses have analyzed studies with percentage of nonoverlap data (PND), nonoverlapping of all pairs (NAP), and Tau-*U* (Ninci et al. 2015; Roth et al. 2014; Wehmeyer et al. 2006). Tau-*U* is an effect size that surpasses PND and NAP by measuring nonoverlap in compared, adjacent phases while correcting for trend (Parker et al. 2011a, b). Tau-*U*'s unique element, accounting for trend, can create a better definition of the level of intervention effectiveness with more statistical power, unlike regression and other nonoverlap methods (Parker et al. 2011b). Though Tau-*U* is a fairly new measure, multiple meta-analyses have used Tau-*U* to assess study outcomes (Bowman-Perrott et al. 2013; Camargo et al. 2014; Hong et al. 2017; Ninci et al. 2015; Soares 2011). Tau-*U* can provide a more thorough analysis of individual and overall outcomes reported in this meta-analysis.

Many of the existing employment curricula and interventions included in transition plans for individuals with DD have not been put under methodological scrutiny. Employment intervention literature is lacking in overall and comparative analyses for participant characteristics, context, independent variables, dependent variables, and effect size measures (Taylor et al. 2012). A meta-analysis can incorporate these elements for a range of studies with more systematic analyses that strengthen conclusions (Lipsey and Wilson 2001). This meta-analysis will be the first to use Tau-*U* effect sizes to measure efficacy of all SCED interventions that meet *WWC* basic design standards (Kratochwill et al. 2010, 2013) and address the teaching of employment skills for individuals with a range of intellectual and developmental disability diagnoses. This meta-analysis considered the following research question in addition to four subcategories:

1. What is the magnitude of effect of employment skill interventions on acquisition and performance of employment skills for individuals with intellectual and other developmental disabilities?

a. Is the investigated intervention more effective with (a) certain types of interventions, (b) certain types or complexities of target skills, (c) certain populations of individuals with disabilities (i.e., participant characteristics), and (d) certain contexts (i.e., setting and implementer characteristics)?

Methods

Search Procedures

Articles for possible inclusion were identified, as of 2015, using electronic searches of the following databases simultaneously: *(a) Academic Search Complete, (b) Applied Technology Full Text*, *(c) ERIC*, *(d) Education Full Text*, *(e) Professional Development Collection*, *(f) Psychology and Behavioral Sciences Collection*, *(g) Social Science Full Text*, *(h) Vocational and Career Collection*, and *(i) Vocational Studies Complete*. There were no specifications for date of publication and the overall search included both peer-reviewed and nonpeer-reviewed articles to combat publication bias, especially the exclusion of dissertations (Cooper 2010). Key search terms were separated into two groups and then combined with each other. Each search term in group 1 was paired with every word from group 2 by the word *and*, creating a Boolean search. The first group contained the words *autis**, *Asperger**, *ASD*, *PDD**, *pervasive developmental disorder*, *development* disab**, *low-incidence dis**, *intellectual* disab**, *mental* retard**, and *multiple disab**. The second group contained the key words *employ**, *career**, *vocation**, *employ* skill**, *career skill**, *vocation* skill**, and *job skill**. An asterisk was added to many search terms for a more encompassing search due to the variations of these words when referred to in published articles.

Inclusion and Exclusion Criteria

Title and Abstract Inclusion/Exclusion

To create a clear objective for the review, a set of inclusion/exclusion criteria was applied to the title and abstract of each article. These criteria consisted of (a) SCED, (b) at least one participant diagnosed with a DD, (c) the dependent variable measures consist of transition skills (i.e., independent living skills, employment/vocational skills, social skills), (d) an intervention component is the independent variable, and (e) a journal article or dissertation published in English. The criterion for the target dependent variables (i.e., transition skills) was broad in this abstract/title screening, even though the focus of this meta-analysis was employment skills, in order to avoid premature exclusion of a source. In addition, even though employment skill interventions were more geared toward adolescent or adults with disabilities, an age criterion was not applied during the title and abstract inclusion/exclusion screening because individuals under the age of 12 could have been involved in pre-employment skills training. If the title and abstract did not include enough information to be confidently included or excluded based on the criteria given above, the entire article was downloaded and the inclusion and exclusion criteria were applied to the full text.

Full-Text Inclusion/Exclusion

After excluding articles based on the information extracted from the titles and abstracts, full-text criteria were applied to all of the remaining articles. The criteria consisted of (a) a SCED (i.e., reversal/withdrawal, alternating treatments, multiple baseline, multiple probe, multi-element design); (b) at least one participant diagnosed with a DD; (c) employment/vocational, or pre-employment/prevocational skills (i.e., employment skills that are taught before the individual is of age to engage in educational transition planning and/or legally attain employment) measured as dependent variables; (d) an intervention implemented to improve and promote independent performance of employment skills; and (e) a line-graph representation of time-series data regarding task completion and independent performance of target skills (e.g., number of prompts needed, percent of steps completed independently). Employment skill–dependent variables and intervention inclusion specifications were applied to focus this review on interventions targeting employment skills. The last criterion, line-graph representation of data, was applied to aid in data extraction visual analysis.

Design Standards

The last round of inclusion and exclusion criteria required the analysis of each experiment (i.e., each single-case example) included in a study using the basic design standards according to *WWC* (Kratochwill et al. 2010, 2013). Studies that met or met with reservations the basic design standards were included in this meta-analysis.

When reviewing the potential articles for inclusion, raters scored each design standard individually and then provided an overall score. The rating system was adopted from Ninci et al. (2015) as modified from Maggin et al. (2013). The modification made by Ninci et al. (2015) included a score of 1 (minimum standards) to those studies that reported IOA generally for at least 20% of sessions (with no specificity per condition) and a score of 2 (quality standards; previously scored a 1 by Maggin et al. 2013) for those studies that reported IOA for at least 20% of each phase, participant, and/or behavior measure. Boles (2015, available online) added a modification for percentage agreement using a score of 1 for a report of at least

80% agreement overall (with no specificity per condition) and a score of 2 for at least 80% IOA agreement per condition. These modifications were made for these design standards to increase specificity when measuring the quality of each study's conditions versus the general quality. The overall quality design scores included meeting the design standard (score of 2), meeting the design standards with reservations (score of 1), or not meeting the design standard (score of 0). Boles (2015, available online) provides the ratings and descriptions of each standard with modifications of Maggin et al. (2013) found in Ninci et al. (2015).

Ancestral Search

Following the design standards screening, an ancestral search was implemented with the previously included articles. An ancestral search included searching through all of the references given in the included studies and running these references through inclusion and exclusion processes. Forty-three studies for possible inclusion were reviewed via the full-text inclusion/exclusion criteria. Ten remaining studies were then analyzed for inclusion based upon the basic design criteria (Kratochwill et al. 2013). A summarized list of the number of articles remaining after every screening process is reported in Fig. 1.

With the completion of all screening processes, a total of 34 studies were categorized as meeting the standards with reservation and five studies were categorized as meeting the design standards. The list of these studies and their design standard scores can be found in Table 1 of Boles et al. (2016). Please note that these design standards apply separately to each experiment (i.e., all single-case designs represented in each study; Kratochwill et al. 2013); therefore, when the design standards did not meet the minimum standards, that experiment's data was excluded, while another participant from the same study (if he or she possesses a data set that meets the minimum standards) was included in the moderator analyses.

Interrater Reliability for Inclusion/Exclusion

Two raters independently screened 30% of the sources accumulated after the database search for the title and abstract and full-text inclusion/exclusion process. The sources were randomly selected from the total accumulation of sources to avoid selection bias. The basic design standards inclusion/exclusion screening was completed by two raters for 50% of the articles. When a disagreement occurred in any of the inclusion and exclusion processes, two raters discussed the discrepancy and reached a consensus. Interrater reliability (IRR) for inclusion/exclusion screenings was calculated via simple percent agreement (total number of agreements divided by agreements plus disagreements and then multiplied by 100) and Cohen's kappa (Cohen 1960). The title/abstract and full-text IRR was scored as 99 and 93% agreement with kappa scores of 0.72 and 0.84, respectively. The basic design standards screening yielded an IRR of 96% agreement and a kappa score of 0.89.

Coding

Descriptive coding and moderator coding based on the characteristics of each study included (a) intervention type, (b) type of target behavior, (c) task complexity, (d) participant age, (e) participant IQ, (f) participant diagnosis, (g) setting, and (h) interventionist. In the case that a participant, intervention, or other characteristic did not fit into the listed levels (e.g., age not provided), the study characteristic was labeled as OTH (i.e., other) and excluded from that specific moderator analysis.

Intervention Type (Independent Variable)

Common interventions for teaching functional living skills, including employment skills, have included video modeling, audio cueing or coaching, visual schedules or scripts, and prompting procedures (Ninci et al. 2015). The first moderator level, video modeling interventions (VM; e.g., video prompting, video priming, video self-modeling, point of view video modeling, adult/peer video modeling, or in vivo modeling), was comprised of a video or live modeling of a target skill by an individual (e.g., the individual, peer, or an adult). After the participant watched the video or live modeling, the target skill was performed. The second level, audio cueing or audio coaching (AC), included pre-recorded or live prompts via the interventionist from a distance while the participant performed the target skill. These audio cues are typically delivered via earpieces or technological device. The third level, visual interventions (VIS), included static pictures/words or scripts individually or in a group of like items that represent the target skill as a whole or each step of the target skill. The last level, prompting interventions (PM), included the systematic use of more or less intrusive prompts to promote independent performance of the target skill.

Studies implementing video modeling interventions contained the largest number of contrasts compared to the other interventions. After consultation with a statistical expert, it was recommended to focus on the intervention with a large number of contrasts to provide more confidence in the results. However, the literature does not state a specific range for the recommended number of contrasts. Therefore, it was decided to focus on the intervention with the largest number of contrasts, by over 50 from the other intervention types, to provide more sound results based on the recommendations given.

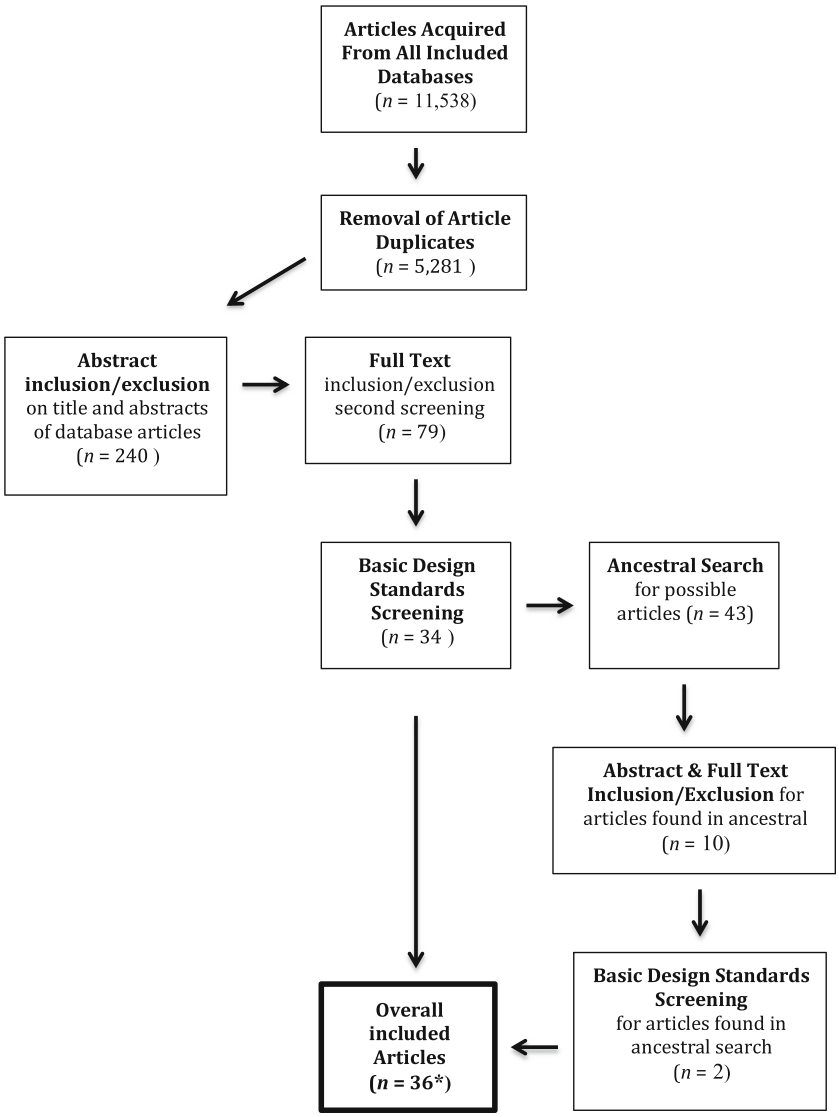


Fig. 1 A summary of the total number of sources and articles remaining after each inclusion and exclusion screening. *Note.* *Three articles included two studies in each article for a total of 39 studies. Only 38 of these studies were unique (i.e., at least one different participant)

Dependent Variables

Each video modeling study was analyzed according to the type of skill targeted by the intervention and the complexity of the skill.

Target Behavior The type of target behavior is important due to the range of skills included in the definition of employment skills. This moderator aids in answering the question: are video modeling interventions just as effective or more effective when teaching different types of target skills? This moderator informs teachers, practitioners, and researchers what skills are best taught with certain interventions. The target behavior moderator separated job-specific skills from employability skills. Job-specific skills are only relevant to the current place of employment, and this skill would be difficult to generalize to another employment setting. Employability skills can generalize across employment contexts because the skill is taught broadly and can be used when performing a variety of job tasks. The target behavior moderator included five levels: SS (i.e., social skills such as requests, interview skills, interacting with customers, asking questions; employability skill), CLEAN (i.e., cleaning the workplace, cleaning a certain product; job-specific skill), ASSEM (i.e., assembly or packaging, stuffing envelopes or folders, putting together more than one piece of an object; job-specific skill), REP (i.e., repetitive tasks that do not require assembly and occur more than once in one sitting such as copying, faxing, or stocking; job-specific skill), and TRANS (i.e., transition from tasks or task steps; employability skill).

Task Complexity Each employment skill has a specific level of complexity according to the amount of steps in the task analysis (e.g., making a copy) or difficulty involved in simply performing the skills (e.g., social skills). This possible moderator gives teachers, practitioners, and researchers insight into the effectiveness of interventions depending on the

complexity of the employment skill being taught. The task complexity moderator included five levels: LOW (i.e., low complexity of 1 to 10 steps involved in the task analysis for a specific task), MOD (i.e., moderate complexity of 11 to 20 steps involved in the task analysis for a specific task), HIGH (i.e., high complexity of 21 or more steps involved in the task analysis for a specific task), SS (i.e., social skill tasks, high complexity), and TRANS (i.e., transition from task to task).

Participant Characteristics

The participant age, diagnosis, and IQ variables were coded to give insight into the effects of participant characteristics on the functional relation between various interventions and employment target skills.

Age The participant age consisted of three variable levels: MS/HS (i.e., middle school and high school students ages 12 to 15), HS (i.e., high school students ages 16 to 21), and ADULT (i.e., ages 22 and over). IDEA (2004) requires transition and employment skill planning to begin no later than the individual's 16th birthday. Due to this requirement, the variable levels include an age group (i.e., MS/HS) that does not require transition planning, but may be granted transition planning by an individualized education program (IEP) team; an age group of students (i.e., HS) at the beginning of transition planning through high school; and an age group of adults (i.e., ADULT) who have graduated high school and are not receiving transition services anymore.

Diagnosis Participant diagnosis gives insight into the effectiveness of employment skill interventions on target skills for specific populations. The interventions can be more effective for individuals diagnosed with ASD than those with ID, or there may be no difference at all. There were three levels for this variable: ASD (i.e., autism, autism spectrum disorder, Asperger's, PDD-NOS), ASDID (i.e., autism spectrum disorder with an intellectual disability), and ID (i.e., intellectual disability).

IQ The last participant characteristic is IQ. This variable was broken up into two levels based on the definitions of severity of the intellectual disability: MILD (i.e., score of 50 or above) or MOD/SEV (i.e., score of 49 or below). The definition of intellectual disability also includes deficits in adaptive behavior (i.e., daily living skills), but due to the lack of adaptive skill assessments included in the relevant studies' participant descriptions, this variable only included reported IQ measures.

Setting

Many school and employment programs used simulated training settings that contained materials and stimuli included in the employment setting due to a lack of resources (e.g., time, funds, and/or employment locations) and/or to promote efficiency (Lattimore et al. 2006; Test et al. 2012). In contrast, employment skill training is often preferred in a natural setting (i.e., job-site, on-the-job, or community-based training) with materials and stimuli that are naturally present in that environment to promote generalization (Lattimore et al. 2006; Test et al. 2012; Walker et al. 2010). Setting was coded in two levels: SIM (i.e., a simulated setting is not in the typical setting where the job task is performed, but does include materials and stimuli similar or identical to those found in the employment setting; e.g., a classroom with items to assemble on each desk, a school office with a fax machine, or a training room with office materials) and NAT (i.e., a natural or employment setting where the skills being taught would typically occur with naturally occurring materials and stimuli; e.g., office building, department store, warehouse).

Interventionist

The interventionist is an important moderator variable due to the fact that some interventionists might be more experienced in implementing video modeling interventions (i.e., more precise treatment fidelity), therefore, making the interventions more effective. This moderator is also important because it is beneficial to know if other interventionists with less training, who are present in the participant's natural environment (i.e., school, community, or place of employment), can implement the video modeling intervention correctly and see benefits in target behaviors while promoting maintenance and generalization of the skills in the natural environment. There were three levels for the interventionist variable: TCHR (teacher or instructor), JC (job coach), and RSCHR (researcher or experimenter).

Interrater Reliability for Coding Moderators

All (100%) of the included studies were independently coded by two raters for the aforementioned moderator descriptions. In the instance of a disagreement, two raters discussed and came to a consensus. IRR was measured according to simple percent agreement. Moderator coding IRR was calculated as 94% agreement.

Raw Data Extraction

Raw data were extracted from each included article's line graphs that contained an intervention targeting employment skills for individuals with a developmental disability and met or met with reservations the basic design standards. Data extraction focused solely on the baseline and intervention phases that were adjacent to each other. Each included data point was rank ordered as suggested by Parker et al. (2011a), starting

with 1 as the lowest data point and going up by one point for the next level including one or more data points. If the target behavior was hypothesized to decrease (i.e., job coach prompting), then the rank order was determined by starting at the highest data point (rank of 1) and increasing as the levels moved downward. For example, in a multiple baseline design including at least three different behaviors and an intervention that is introduced to these behaviors at different points in time ($A_1B_1A_2B_2A_3B_3$), each A and B phase pair would be rank ordered separately. The data extraction process included taking a screenshot of the included graph, pasting this screenshot in a spreadsheet file, electronically drawing straight lines that match the level of each included data point, analyzing the rank order of each included data point, and recording the list of rank orders for each included phase.

Calculating Effect Sizes

Tau-*U* is an effect size based on the nonoverlap of data in phase contrasts while accounting for trend in baseline, if needed (Parker et al. 2011b). Tau-*U* measures range from − 1.0 (indicating a reverse of hypothesized effects) to 1.0 (indicating no overlap between phases and 100% improvement in the intervention phase compared to baseline; Parker et al. 2011b). Tau-*U* was calculated for each phase contrast according to each indicated moderator via a Tau-*U* calculator (Davis and Davis 2014). In addition to calculating Tau-*U* and to create a more conservative analysis of the moderator variables, the Kruskal–Wallis one-way analysis of variance was conducted to measure statistical significance (Kruskal and Wallis 1952) or Wilcoxon statistics (Wilcoxon et al. 1963) for comparison of variables with only two levels. If the Kruskal–Wallis measure showed statistical significance, the Dunn post hoc test was implemented to analyze existing pairwise comparisons (Dunn 1964). The only exceptions were the IQ and setting moderators with only two levels per moderator; therefore, it was necessary to run the Wilcoxon two-sample test for significance instead of the Kruskal–Wallis (Wilcoxon et al. 1963).

Interrater Reliability for Data Extraction

All (100%) included graphs were independently rank ordered and visually analyzed by two raters. IRR was attained for all included graphs due to the high possibility of error from inattention to minute details or difficulty analyzing the graphs because of distortion due to poor publication resolution. IRR was measured according to simple percent agreement. In the case of any disagreements in the data extraction process, two raters discussed the discrepancy and reached a consensus. Data extraction IRR yielded 97% agreement.

Results

This meta-analytic review analyzed 235 AB contrasts from 39 studies with a total of 120 participants. Descriptions of strong (i.e., 0.85–0.95), moderate (i.e., 0.70–0.84), or weak effects (i.e., below 0.70) were used to categorize Tau-*U* effect sizes for all items discussed below (Ninci et al. 2015). Tau-*U* scores ranged from 0.34 to 1.00 for each study and − 0.67 to 1.00 for each contrast. A total of 29 studies and over 180 contrasts resulted in independent Tau-*U* scores of 0.85 or above, resulting in strong intervention effects on acquisition and performance of employment skills. Due to the wide range of study components (e.g., participant characteristics, intervention type, and target behaviors), a Tau-*U* moderator analysis and significance testing ($p < 0.05$) were conducted for each intervention type to determine which studies would be included in the remaining moderator analyses. Tables 1, 2, 3, 4, and 5 provide Tau-*U* scores and significance scores (if applicable) for each moderator level given below.

Intervention Type

Tau-*U* scores for the type of intervention implemented ranged from moderate effects of 0.83 CI_{95} [0.79, 0.87] for video modeling interventions to strong effects of 0.93 CI_{95} [0.84, 1.00] for prompting interventions, 0.97 CI_{95} [0.91, 1.00] for visual interventions, and 0.97 CI_{95} [0.85, 1.00] for audio cueing/coaching interventions. The Kruskal–Wallis test did indicate a significant difference between moderator levels ($p = 0.0001$). The Dunn post hoc test indicated a pairwise significant difference between the efficacy of video modeling and audio cueing/coaching interventions. Audio cueing/coaching interventions were shown to have stronger effects than video modeling interventions, but audio cueing interventions only included 4 studies and a total of 17 A–B contrasts; therefore, these results should be interpreted with caution. There were no other significant differences between the other interventions. All Tau-*U* scores and Dunn post hoc results pertinent to the intervention moderator analysis are reported in Table 1.

Studies implementing video modeling interventions yielded the highest number of contrasts ($n = 112$): more than double the contrasts yielded by studies implementing visual interventions ($n = 51$). Audio cueing/coaching, visual, and prompting interventions yielded small numbers of contrasts, making it difficult to break down overall effect sizes by additional moderators. Studies implementing video modeling interventions not only contained enough contrasts to run additional moderator analyses, but the overall moderate Tau-*U* scores calculated for this intervention warranted further analysis. In addition, the sole inclusion of video modeling intervention data in the remaining

Table 1 Tau-*U* scores and significant differences for intervention-type moderator

Intervention Tau-*U* scores				
Intervention type	Number of studies	Number of participants	Number of contrasts	Tau-*U* [CI_{95}]
VM (video modeling)	15	44	112	0.83 [0.79, 0.87]
AC (audio cueing/coaching)	4	13	17	0.97 [0.85, 1.00]
VIS (visuals)	8	31	51	0.97 [0.91, 1.00]
PRMT (prompting)	5	14	28	0.93 [0.84, 1.00]
Intervention Dunn post hoc results				
Group comparisons	Difference in average ranks	Cutoff at alpha = 0.05		Significant differences = **
AC–PRMT	18.83	48.82		
AC–VIS	30.11	44.47		
AC–VM	52.37	41.33		**
PRMT–VIS	11.28	37.35		
PRMT–VM	33.54	33.55		
VIS–VM	22.26	26.82		

moderator analyses was also due to conceptual differences between all four interventions and the need to provide significant results regarding the efficacy of a more commonly implemented intervention's components. For all studies that implemented video modeling interventions, moderator analyses for types of target behaviors, complexity of employment tasks, characteristics of participants, types of settings, and characteristics of interventionists were conducted with studies that implemented video modeling interventions. Significant differences between the moderator levels were conducted with Kruskal–Wallis, Wilcoxon two-sample for moderators with only two levels, and Dunn post hoc tests for moderators with three or more levels.

Dependent Variable

Target Behaviors

Tau-*U* scores for specific target behaviors ranged from weak effects for job-specific repetitive target behaviors to strong effects for job-specific assembly and cleaning behaviors as well as both employability skills (i.e., social skills and transition skills; see Table 2). Job-specific cleaning behaviors yielded the strongest effect 0.92 [0.76, 1.00], but results should be interpreted with caution due to the small number ($n = 5$) of contrasts. Job-specific assembly skills (0.89 CI_{95} [0.80, 0.98]) and employability social skills (0.85 CI_{95} [0.78, 0.92]) yielded strong effects

Table 2 Video modeling: Tau-*U* scores and significant differences for target behaviors

Target behavior Tau-*U* scores				
Target behaviors	Number of studies	Number of participants	Number of contrasts	Tau-*U* [CI_{95}]
REP (repetitive)	5	15	36	0.69 [0.61, 0.78]
ASSEM (assembly)	5	12	21	0.89 [0.80, 0.98]
CLEAN (cleaning)	2	3	5	0.92 [0.76, 1.00]
SS (social skills)	6	17	31	0.85 [0.78, 0.92]
TRANS (transition between tasks/steps)	2	5	16	0.88 [0.75, 1.00]
Target behavior Dunn post hoc results				
Group comparisons	Difference in average ranks	Cutoff at alpha = 0.05	Significant differences	
ASSEM–CLEAN	3.67	44.15		
ASSEM–REP	27.04	24.36	**	
ASSEM–SS	9.17	25.08		
ASSEM–TRANS	0.40	29.44		
CLEAN–REP	30.71	42.35		
CLEAN–SS	12.84	42.76		
CLEAN–TRANS	4.07	45.46		
REP–SS	17.87	21.74		
REP–TRANS	26.64	26.66		
SS–TRANS	8.7732	27.3133		

Table 3 Video modeling: Tau-*U* scores and significant differences for task complexity

Task complexity Tau-*U* scores				
Task complexity	Number of studies	Number of participants	Number contrasts	Tau-*U* [CI_{95}]
LOW (1–10 steps)	4	11	53	0.84 [0.73, 0.95]
MOD (11–20 steps)	4	12	19	0.71 [0.62, 0.80]
HIGH (21 steps and over)	1	4	4	0.77 [0.52, 1.00]
SOC (complex social skills)	6	17	31	0.85 [0.78, 0.92]
TRANS (lower complexity transition skills)	2	5	16	0.88 [0.75, 1.00]
Task complexity Dunn post hoc results				
Group comparisons	Difference in average	Cutoff at alpha = 0.05	Significant differences = **	
HIGH–LOW	17.95	41.67		
HIGH–MOD	5.93	42.57		
HIGH–SOC	19.00	41.11		
HIGH–TRANS	26.06	43.26		
LOW–MOD	23.87	23.55	**	
LOW–SOC	1.05	20.80		
LOW–TRANS	8.12	24.78		
MOD–SOC	24.92	22.55	**	
MOD–TRANS	31.99	26.26	**	
SOC–TRANS	7.07	23.82		

from 21 to 31 contrasts, respectively. Overall, assembly and social behaviors yielded strong effects with narrow confidence intervals and a larger number of contrasts per level; therefore, these results can be interpreted with more confidence compared to the small number ($n = 5$, 16) of contrasts for cleaning and transition behaviors.

The Kruskal–Wallis did indicate a significant difference between target behavior moderator levels ($p = 0.0023$). The Dunn post hoc yielded a pairwise significant difference between the efficacy of video modeling interventions for assembly versus repetitive target behaviors. Assembly behaviors were shown to have stronger effects than repetitive behaviors (see Table 2).

Task Complexity

Tau-*U* scores for task complexity ranged from moderate effects of 0.71 CI_{95} [0.62, 0.80] for moderately complex target behaviors to strong effects of 0.88 CI_{95} [0.75, 1.00] for lower complexity transition skills (see Table 3). Complex social skills and lower complexity transition skills yielded the strongest effects, but transition tasks only included 16 contrasts compared to 31 contrasts for social skill tasks; therefore, these data should be interpreted with caution. Tasks with the lowest complexity included 53 contrasts and yielded a moderate Tau-*U* score of 0.84 CI_{95} [0.73, 0.95] indicating a more precise measure of the efficacy of video modeling interventions when

Table 4 Video modeling: Tau-*U* scores for participant characteristic moderators

	Number of studies	Number of participants	Number of contrasts	Tau-*U* [CI_{95}]
Age				
MS/HS (12–15 years)	3	7	31	0.85 [0.75, 0.96]
HS (16–21 years)	9	23	57	0.82 [0.77, 0.88]
ADULT (22 years and over)	7	14	24	0.83 [0.75, 0.90]
IQ				
MILD (50 or above)	9	16	43	0.87 [0.80, 0.94]
MOD/SEV (below 50)	9	15	44	0.92 [0.85, 0.99]
Diagnosis				
ASD (autism spectrum disorder)	3	10	10	0.79 [0.68, 0.89]
ASDID (ASD + ID)	5	12	45	0.81 [0.73, 0.90]
ID (intellectual disability)	9	22	57	0.85 [0.80, 0.91]

teaching lower complexity tasks to individuals with developmental disabilities.

The Kruskal–Wallis did indicate a significant difference between task complexity moderator levels ($p = 0.0021$). The Dunn post hoc yielded a pairwise significant difference between the efficacies of video modeling interventions for low versus moderate complexity tasks, moderate versus high complexity social tasks, and moderate versus low complexity transition tasks. Video modeling interventions targeting low complexity tasks, high complexity social tasks, and low complexity transition tasks had significantly stronger effects compared to their counterparts (see Table 3).

Participant Characteristics

Three moderator variables focusing on specific participant characteristics (i.e., age, IQ, and diagnosis) were analyzed using Tau-*U* effect sizes for each level. Each variable's number of studies, number of participants, number of analyses, and Tau-*U* scores are reported in Table 4.

Age

Tau-*U* scores for video modeling interventions with participants differentiated by age group ranged from moderate effects of 0.82 CI_{95} [0.77, 0.88] for participants ages 16 to 21 years old and 0.83 CI_{95} [0.75, 0.90] for those 22 years old or older to strong effects of 0.85 CI_{95} [0.75, 0.96] for those 12 to 15 years old (see Table 4). The Kruskal–Wallis and Dunn post hoc tests did not show any significant difference between levels ($p = 0.1415$).

IQ

For participants categorized with mild (i.e., 50 or above) and moderate to severe IQs (i.e., below 50), both IQ ranges yielded strong Tau-*U* scores of 0.87 CI_{95} [0.80, 0.94] and 0.92 CI_{95} [0.85, 0.99], respectively. Mild and moderate to severe IQ variable levels included a similar number of contrasts ($n = 43, 44$), which creates a more precise comparison of effect sizes (see Table 4). The Wilcoxon two-sample test did not show any significant difference between IQ levels ($p = 0.2620$).

Diagnosis

Tau-*U* scores for video modeling interventions that were implemented with participants with specific diagnoses ranged from moderate effects of 0.79 CI_{95} [0.68, 0.89] for participants with ASD and 0.81 CI_{95} [0.73, 0.90] for participants with ASD with ID to strong effects of 0.85 CI_{95} [0.80, 0.91] for participants with ID. Individuals with comorbid ASD with ID and individuals with ID included the largest number of contrasts ($n = 45, 57$); therefore, the strong effects for individuals with ASD with ID and moderate effects for individuals with ID can be interpreted more precisely compared to individuals with ASD ($n = 10$ contrasts). The Kruskal–Wallis and Dunn post hoc tests did not show a significant difference between the participants in each diagnostic category ($p = 0.3528$).

Setting

Video modeling interventions implemented in natural settings yielded moderate effects of 0.81 CI_{95} [0.75, 0.87] and interventions implemented in simulated settings yielded strong effects of 0.85 CI_{95} [0.79, 0.91]. Both setting moderator levels included more than 40 contrasts each; therefore, more precise conclusions could be made based on outcomes (see Table 5). The Wilcoxon two-sample test did not show any significant difference between setting moderator levels ($p = 0.1016$).

Interventionist

Tau-*U* scores for interventionist variables ranged from weak effects of 0.65 CI_{95} [0.79, 0.92] for implementation by job coaches to moderate effects of 0.78 CI_{95} [0.64, 0.92] for implementation by teachers and to strong effects of 0.92 CI_{95} [0.79, 1.00] for implementation by researchers (see Table 5). Due to the lack of information given to describe the interventionists in each study, the number of contrasts ($n = 12, 14$) for job coach and researcher variables was too small to break down effect sizes for each level. Therefore, the Tau-*U* scores gleaned for each interventionist moderator level are inconclusive.

Discussion

This meta-analysis analyzed the outcomes of 39 SCED studies that implemented a range of interventions to promote acquisition and independence in performing employment-related behaviors for individuals with developmental disabilities. This meta-analysis contributes to the literature regarding employment skills and is unique in its focus, statistical methodology, and analysis of relevant variables. The results of this meta-analysis provide evidence of moderate to strong effects across all employment skill interventions. These results corroborate the outcomes of Ninci et al. (2015) regarding employment skill interventions for a sample of articles that only included individuals with ASD. Furthermore, this study adds to the previous reviews (e.g., Gilson et al. 2017; Hedley et al. 2017) by conducting comparative analyses for different moderators including participant characteristics, context, independent variables, dependent variables, and effect size measures,

Table 5 Video modeling: Tau-*U* scores for setting and interventionist moderators

	Number of studies	Number of participants	Number of contrasts	Tau-*U* [CI_{95}]
Setting				
NAT (natural)	8	24	44	0.81 [0.75, 0.87]
SIM (simulation)	8	24	68	0.85 [0.79, 0.91]
Interventionist				
JC (job coach)	1	4	12	0.65 [0.52, 0.79]
RSCHR (researcher)	2	6	14	0.92 [0.79, 1.00]
TCHR (teacher)	1	3	21	0.78 [0.64, 0.92]

which leads to more systematic analyses that strengthen the conclusions of its findings.

The first question considered in this meta-analysis inquired about the difference in effects across intervention types. Strong effects were found for three out of the four interventions (i.e., audio cueing/coaching, visuals, and prompting). Moderate effects resulted from video modeling interventions, which yielded overall lower effect sizes than audio cueing/coaching, visual, and prompting interventions. Despite lower effects, video modeling interventions resulted in moderate effects that are consistent with prior work that also categorized video modeling as a quality, effective, and evidence-based practice for individuals with developmental disabilities (Bellini and Akullian 2007; Boles et al. 2016; Mason et al. 2012, 2013). According to Boles et al. (2016); visual, audio cueing, and prompting interventions are not considered evidence-based interventions to teach employment skills to individuals due to the lack of 20 experiments, 5 studies, or 3 different research groups. These interventions do not meet the 5-3-20 rule supported by Horner et al. (2005) and Kratochwill et al. (2010, 2013). More research is needed to deem these interventions effective and evidence-based.

The second question considered in this meta-analysis inquired about the difference in video modeling intervention effects across the type and complexity of target behaviors. Video modeling interventions targeted job-specific skills, including (a) repetitive, (b) assembly, and (c) cleaning, and employability skills, including (a) social and (b) transition skills, all of which yielded moderate to strong effects across target behaviors. Many of the behaviors and video models of these behaviors were constructed with thorough task analyses, which may contribute to the success of the participants in accomplishing these tasks. Video modeling yielded weak effects (i.e., 0.69) for repetitive behaviors. This may be due to the diversity of tasks included in repetitive behaviors (i.e., copying, faxing, stocking), which may account for the variability of outcomes seen in the high number of contrasts analyzed for these behaviors. There was a pairwise significant difference in effects between job-specific assembly and repetitive tasks. The strong effects seen in assembly tasks may be due to less diversity of task complexity (i.e., low to moderate complexity) than repetitive behaviors (i.e., low, moderate, and high complexity), which may have resulted in better participant performance with assembly tasks.

Social skill behaviors yielded strong effects and more than 30 contrasts were analyzed for this skill. This finding may give more insight into the effectiveness of video modeling in teaching a diverse set of social skills (i.e., interacting with customers, interacting with employers, and interacting with others during an interview) and other less structured employability skills. Social skills are complex and very difficult to teach in a variety of contexts; so, more research should be done to verify the importance of video modeling interventions in teaching social skills. Overall, these strong effects across a diverse set of job-specific and employability skills are promising for individualizing interventions based on individuals' needs for employment skill acquisition.

In addition to the effects of interventions targeting certain types of employment skills, all complexity levels of these skills yielded moderate to strong effects. Video modeling interventions have been determined by previous studies as effective in teaching individuals with ASD and other developmental disabilities a variety of functional living skills (i.e., independent living, social, and employment skills), which support the moderate to strong effects across the range of skills and task complexities (Mason et al. 2012, 2013; Ninci et al. 2015). The significant differences between task complexities may be due to the small number of individuals from each diagnostic category who were taught at each level of task complexity. There were not enough participants in each diagnostic category for each level of complexity to determine if participant diagnosis was related to the acquisition and performance of certain tasks. Also, the significant difference could have resulted in the use of different video modeling components to teach these skills (e.g., video prompting versus video priming), which could make a difference in the acquisition and performance of designated task analyses. Overall, these results indicate that skills requiring a range of skill levels can be taught, acquired, and performed as a result of video modeling interventions. The range of complexity allows for

researchers and practitioners to implement more individualized video modeling interventions for individuals with developmental disabilities with a variety of deficits in employment skills.

The third question considered in this meta-analysis inquired about the difference in video modeling intervention effects across participant characteristics (i.e., age, diagnosis, and IQ). Moderate and strong effects were calculated across participants ages 12 years old and older (see Table 4). A majority of studies implemented video modeling interventions for participants ages 16 to 21 years old, resulting in moderate intervention effects. These moderate effects for individuals 16 years and older corroborates the IDEA (2004) federal mandate to begin transition planning no later than age 16. IDEA (2004) also mandates that transition planning can begin whenever the IEP team determines it to be appropriate. Video modeling interventions implemented with individuals ages 12 to 15 years old yielded the strongest effects (0.85) compared to other age groups. With many special education stakeholders calling for transition planning to begin before the age of 16, the efficacy of video modeling interventions with these younger age groups is promising in teaching employment or pre-employment skills. If transition planning begins at age 12, this leaves up to 9 years of employment skill instruction before these individuals graduate high school (Johnson 2012). This may allow for more preparation for individuals to acquire employment and successfully fulfill the duties of employment.

Strong effects were also seen for individuals across IQ scores and participants diagnosed with ID. No significant differences existed between individuals with ASD, ASD with ID, and ID, or individuals with mild and moderate to severe IQ scores, which indicates that moderate to strong video modeling intervention effects can occur for individuals with a range of developmental disabilities and intellectual functioning levels (see Table 4). The lack of significant differences within IQ and diagnosis variable levels may be due to the inability to assess detailed similarities and differences between these levels. For example, cognitive functioning is typically related to IQ and adaptive skills, but only a small portion of the included studies reported adaptive assessment scores, making it difficult to assess which component of cognitive functioning may influence intervention effects. More research is needed in implementing employment skill interventions with a diverse population of individuals with DD to assess the detailed components of variables and their influence on intervention effects. Overall, these IQ and diagnosis results mirror previous reviews and meta-analyses outcomes determining the efficacy of video modeling interventions for individuals with ASD and other DD (Bellini and Akullian 2007; Mason et al. 2012, 2013).

The final question considered in this meta-analysis inquired about the difference in video modeling effects across settings and interventionist characteristics. Video modeling interventions implemented in both natural and simulated settings yielded moderate effects in the natural setting and strong effects in the simulated setting with no significant difference in efficacy. Similar effect sizes for these settings may be due to the fact that simulated settings replicate the materials, the individuals present, and/or the environmental arrangement of the setting in which the employment skill would typically take place in (i.e., natural setting; Lattimore et al. 2006; Test et al. 2012; Walker et al. 2010). These effect sizes are promising for the practicality and efficacy of interventions implemented in a simulated setting if implementation in a natural setting is not plausible. In many cases, there are limited resources (i.e., lack of time, facilities, job coaches) when implementing employment skill interventions, and in some cases, it is not practical to teach these skills in a busy and overstimulating work environment. These results should be interpreted with caution due to the absence of generalization analyses in this meta-analysis.

In addition to the setting, video modeling intervention effects were inconclusive in the analyses for interventionist characteristics. The lack of reporting interventionist characteristics in many studies led to the small number of contrasts for researchers and job coaches. Also, the larger effect size yielded by researchers performing as interventionists may be due to additional experience or training regarding the procedures and purpose of the experiment. When the sole interventionist is a researcher, there is no way to measure the efficacy of an intervention when implemented by an individual (e.g., job coach, coworker) that typically works with the individual with a disability. Therefore, future research should focus on thoroughly reporting interventionist characteristics and video modeling implementation by a variety of individuals present in the everyday training or work environment of the individual with a disability (i.e., teachers, job coaches, and coworkers, employers) to promote generalization of skills and maintenance of intervention implementation in the natural environment (Likins et al. 1989). If research replicates video modeling or other employment skill interventions with a variety of interventionists, effect size analyses can produce more precise conclusions in the future.

Implication for Practice

This meta-analysis thoroughly analyzed relevant data and specific moderators to inform researchers and practitioners of employment skill interventions' effectiveness in multiple contexts. Video modeling techniques, in general, have been defined as portable, efficient, and socially acceptable interventions that can be beneficial to adolescent and adult individuals with developmental disabilities across contexts (Mason et al. 2012, 2013). This meta-analysis applied measures of efficacy to video modeling interventions that focused on teaching employment skills. Video modeling interventions were seen to be

effective across a majority of target behaviors, task complexities, participant age groups, participant IQ levels, participant developmental disability diagnoses, intervention settings, and interventionist characteristics. As video modeling is a mobile intervention that can be practically applied by the individual with a disability, it is promising that this intervention can be effectively used in the work place to promote independence and maintenance of acquiring and performing employment skills (Bellini and Akullian 2007; Mason et al. 2012, 2013). All outcomes provide pertinent information regarding job acquisition and maintenance for all stakeholders (e.g., teachers, individuals with DD, job coaches, employers) to decrease rates of unemployment for individuals with DD (Newman et al. 2009, 2011; Sanford et al. 2011). Overall, the results of the analyses in the current review provided evidence for effective interventions to teach individuals with DD a variety of employment skills. This is promising when addressing issues surrounding the transition processes in special education.

Limitations and Implications for Future Research

Overall, a majority of interventions and moderators yielded strong effects, but there were three limitations in this meta-analysis. First, there were studies or contrasts that included small sample sizes for some variable levels (i.e., participants 12 to 15 years old, diagnosis of ASD, audio cueing/coaching and prompting interventions, researcher interventionists, and transition skills). This is not surprising considering that more than half of the studies identified through the literature searches were excluded in the initial stage of the analysis given that those articles did not meet the preset design standards (Kratochwill et al. 2010, 2013). Most of the studies that failed to meet the design standards were mainly due to a lack of specificity of IOA measures or collected only a small number of data points per phase, leading to the exclusion of those studies from this review. Future research should therefore strive to conduct studies with high methodological quality that meet current standards for SCED.

Regarding the sample size, the small number of studies or contrasts per level may have inflated the strong effect sizes because these samples only included highly positive outcomes, which may not be representative of all intervention or moderator effects. This is also apparent in the presence of larger confidence intervals due to the lack of precise measurement for smaller samples. Small samples in these areas indicate the need for more employment skill research for different interventions, ages, diagnoses, and employment-related skills. In SCEDs, it is recommended to have at least five studies with a minimum of 20 participants to assure an adequate power to detect statistical power while ruling out the possibility of inflating or deflating effects sizes due to a small sample size in effect size calculation (Kratochwill et al. 2010, 2013). However, it is not always possible to control a sample size when synthesizing the literature. Considering the complexity and nature of SCED, therefore, it is suggested that future analyses should include multiple statistical metrics along with a comparison between effect sizes and independent visual analysis of data to ensure the validity for meta-analytic reviews in SCEDs (Brossart et al. 2014; Burns 2012; Maggin and Odom 2014; Whalon et al. 2015).

Second, data extraction and effect size analyses did not include maintenance or generalization outcomes. Maintenance and generalization data are important to the continuation of study effects whether in the same or different contexts. Future analyses of these phases are needed to provide more information on the continuation of treatment effects on employment skills.

Third, this meta-analysis focused solely on SCEDs implementing employment skill interventions. Group studies were not considered in these analyses due to the difficulty of comparing two different effects sizes from SCED and group experimental design. Group studies should be analyzed in future reviews to offer additional insights into overall employment skill intervention effects.

Compliance with Ethical Standards

Conflict of Interest The authors declare that they have no conflict of interest.

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References

*Indicates studies included in the review

*Allen, K.D., Wallace, D.P., Greene, D.J., Bowen, S.L., & Burke, R.V. (2010a). Community-based vocational instruction using videotaped modeling for young adults with autism spectrum disorders performing in air-inflated mascots. Focus on Autism and Other Developmental Disabilities, 25, 186–192. https://doi.org/10.1177/1088357610377318.

*Allen, K.D., Wallace, D.P, Renes, D., Bowen, S.L., & Burke, R.V. (2010b). Use of video modeling to teach vocational skills to adolescents and young adults with autism spectrum disorders. Education and Treatment of Children, 33, 339–349. https://doi.org/10.1353/etc.0.0101.

*Allen, K.D., Burke, R.V., Howard, M.R., Wallace, D.P., & Bowen, S.L. (2012). Use of audio cuing to expand employment opportunities for adolescents with autism spectrum disorders and intellectual disabilities. Journal of Autism and Developmental Disorders, 42, 2410–2419. https://doi.org/10.1007/s10803-012-1519-7.

Alwell, M., & Cobb, B. (2009). Functional life skills curricular interventions for youth with disabilities: systematic review. *Career Development for Exceptional Individuals, 32*, 82–93. https://doi.org/10.1177/0885728809336656.

Banda, D. R., Dogoe, M. S., & Matuszny, R. M. (2011). Review of video prompting studies with persons with developmental disabilities.

Education and Training in Autism and Developmental Disabilities, 46, 514–527.

Bellini, S., & Akullian, J. (2007). A meta-analysis of video modeling and self-modeling interventions for children and adolescents with autism spectrum disorders. *Exceptional Children, 73*, 264–297. https://doi.org/10.1177/001440290707300301.

*Bennett, K., Brady, M.P, Scott, J., Dukes, C., & Frain, M. (2010). The effects of covert audio coaching on the job performance of supported employees. Focus on Autism and Other Developmental Disabilities, 25, 173–185. https://doi.org/10.1177/1088357610371636.

*Bennett, K.D., Rangasamy, R., & Honsberger, T. (2013a). The effects of covert audio coaching on teaching clerical skills to adolescents with autism spectrum disorder. Journal of Autism and Developmental Disorders, 43, 585–593. https://doi.org/10.1007/s10803-012-1597.

*Bennett, K.D., Rangasamy, R., & Honsberger, T. (2013b). Further examination of covert audio coaching on improving employment skills among secondary students with autism. Journal of Behavioral Education, 22, 103–119. https://doi.org/10.1007/s10864-013-9168-2.

*Bereznak, S., Ayres, K.M., Mechling, L.C., & Alexander, J.L. (2012). Video self-prompting and mobile technology to increase daily living and vocational independence for students with autism spectrum disorders. Journal of Developmental and Physical Disabilities, 24, 269–285. https://doi.org/10.1007/s10882-012-9270-8.

Billstedt, E., Gillberg, C., & Gillberg, C. (2005). Autism after adolescence: population-based 13- to 22-year follow-up study of 120 individuals with autism diagnosed in childhood. *Journal of Autism and Developmental Disorders, 35*, 351–360. https://doi.org/10.1007/s10803-005-3302-5.

Boles, M. (2015). Basic quality design standards rating scale for single-case design. Retrieved September 1, 2015, from http://hdl.handle.net/1969.1/153769 .

Boles, M. B., Ganz, J. B., Hagan-Burke, S., Gregori, E. V., Neely, L., Mason, R. A., Zhang, D., & Willson, V. (2016). Quality review of single-case studies concerning employment skill interventions for individuals with developmental disabilities. *Cadernos de Educacão (UFPel), 53*, 15–51. Retrieved Jan 28, 2019, from https://periodicos.ufpel.edu.br/ojs2/index.php/caduc/article/view/8861/9194.

Bowman-Perrott, L., Davis, H., Vannest, K., Williams, L., Greenwood, C., & Parker, R. (2013). Academic benefits of peer tutoring: a meta-analytic review of single-case research. *School Psychology Review, 4*, 39–55.

Brossart, D. F., Vannest, K. J., Davis, J. L., & Patience, M. A. (2014). Incorporating nonoverlap indices with visual analysis for quantifying intervention effectiveness in single-case experimental designs. *Neuropsychological Rehabilitation, 24*, 464–491. https://doi.org/10.1080/09602011.2013.868361.

Burns, M. K. (2012). Meta-analysis of single-case design research: introduction to the special issue. *Journal of Behavioral Education, 21*, 175–184. https://doi.org/10.1007/s10864-012-9158-9.

Camargo, S. P. H, Rispoli, M., Ganz, J., Hong E., Davis, H., & Mason, R. (2014). A review of the quality of behaviorally-based intervention research to improve social interaction skills of children with ASD in inclusive settings. *Journal of Autism and Developmental Disorders, 44*, 2096–2116.

Carter, E. W., Austin, D., & Trainor, A. A. (2012). Predictors of post-school employment outcomes for young adults with severe disabilities. *Journal of Disability Policy Studies, 23*, 50–63. https://doi.org/10.1177/1044207311414680.

*Cavkaytar, A. (2012). Teaching café waiter skills to adults with intellectual disability: a real setting study. Education and Training in Autism and Developmental Disabilities, 47, 426–437.

*Chandler, W., Schuster, J.W., & Stevens, K.B. (1993). Teaching employment skills to adolescents with mild and moderate disabilities using constant time delay procedure. Education and Training in Mental Retardation, 28, 155–168.

*Chang, Y.J., Kang, Y.S., & Huang, P.C. (2013). An augmented reality (AR)-based vocational task prompting system for people with cognitive impairments. Research in Developmental Disabilities, 34, 3049–3056. https://doi.org/10.1016/j.ridd.2013.06.026.

*Cihak, D.F., Alberto, P.A., Kessler, K.B., & Taber, T.A. (2004). An investigation of instructional scheduling arrangements for community-based instruction. Research in Developmental Disabilities, 25, 67–88. https://doi.org/10.1016/j.ridd.2003.04.006.

*Cihak, D.F., Kessler, K.B., & Alberto, P.A. (2007). Generalized use of a handheld prompting system. Research in Developmental Disabilities, 28, 397–408. https://doi.org/10.1016/j.ridd.2006.05.003.

*Cihak, D.F., Kessler, K., & Alberto, P.A. (2008). Use of a handheld prompting system to transition independently through vocational tasks for students with moderate and severe intellectual disabilities. Education and Training in Developmental Disabilities, 43, 102–110.

Cohen, J. (1960). A coefficient of agreement for nominal scales. *Educational and Psychological Measurement, 20*, 37–46. https://doi.org/10.1177/001316446002000104.

*Connis, R.T. (1979). The effects of sequential pictorial cues, self-recording, and praise on the job task sequencing of retarded adults. Journal of Applied Behavioral Analysis, 12, 355–361. https://doi.org/10.1901/jaba.1979.12-355.

Cooper, H. (2010). *Research synthesis and meta-analysis: a step-by-step approach* (4th ed., Applied social research methods series, vol. 2). Thousand Oaks, CA: Sage.

Council for Exceptional Children. (2014). *Council for Exceptional Children standards for evidence-based practices in special education.* Retrieved August 1, 2015, from http://www.cec.sped.org/Standards/Evidence-Based-Practice-Resources-Original .

Davis, J. L., & Davis, J. H. (2014). *Meta-analytical software for single case research*. College Station: Author.

*Devlin, P. (2008). Enhancing the job performance of employees with disabilities using the self-determined career development model. Education and Training in Developmental Disabilities, 43, 502–513.

*DiPipi-Hoy, C., Jitendra, A.K., & Kern, L. (2009). Effects of time management instruction on adolescents' ability to self-manage time in a vocational setting. The Journal of Special Education, 43, 145–159. https://doi.org/10.1177/0022466908317791.

Dolyniuk, C. A., Kamens, M. W., Corman, H., DiNardo, P. O., Totaro, R. M., & Rockoff, J. C. (2002). Students with developmental disabilities go to college: description of a collaborative transition project on a regular college campus. *Focus on Autism and Other Developmental Disabilities, 17*, 237–241. https://doi.org/10.1177/10883576020170040601.

*Dotto-Fojut, K.M., Reeve, K.F., Townsend, D.B., & Progar, P.R. (2011). Teaching adolescents with autism to describe a problem and request assistance during simulated vocational tasks. Research in Autism Spectrum Disorders, 5, 826–833. https://doi.org/10.1016/j.rasd.2010.09.12.

Dunn, O. J. (1964). Multiple contrasts using rank sums. *Technometrics, 6*, 241–252. https://doi.org/10.2307/1266041.

Gast, D. L., & Ledford, J. R. (Eds.). (2014). *Single case research methodology: applications in special education and behavior sciences* (2nd ed.). New York, NY: Routledge.

Gilson, C. B., Carter, E. W., & Biggs, E. E. (2017). Systematic review of instructional methods to teach employment skills to secondary students with intellectual and developmental disabilities. *Research and Practice for Persons with Severe Disabilities, 42*, 89–107. https://doi.org/10.1177/1540796917698831.

*Goh, A.E., & Bambara, L.M. (2013). Video self-modeling: a job skills intervention with individuals with intellectual disability in employment settings. Education and Training in Autism and Developmental Disabilities, 48, 103–119.

Grigal, M., & Deschamps, A. (2012). Transition education for adolescents with intellectual disabiltiy. In M. L. Wehmeyer & K. W. Webb (Eds.), *Handbook of adolescent transition education for youth with disabilities* (pp. 398–416). New York, NY: Routledge.

Hanley-Maxwell, C., & Izzo, M. V. (2012). Preparing students for the 21st century workforce. In M. L. Wehmeyer & K. W. Webb (Eds.), *Handbook of adolescent transition education for youth with disabilities* (pp. 139–155). New York, NY: Routledge.

Hedley, D., Uljarević, M., Cameron, L., Halder, S., Richdale, A., & Dissanayake, C. (2017). Employment programmes and interventions targeting adults with autism spectrum disorder: a systematic review of the literature. *Autism, 21*, 929–941. https://doi.org/10.1177/1362361316661855.

Hendricks, D. R. (2010). Employment and adults with autism spectrum disorders: challenges and strategies for success. *Journal of Vocational Rehabilitation, 32*, 125–134. https://doi.org/10.3233/JVR-2010-0502.

Hendricks, D. R., & Wehman, P. (2009). Transition from school to adulthood for youth with autism spectrum disorders. *Focus on Autism and Other Developmental Disabilities, 24*, 77–88. https://doi.org/10.1177/1088357608329827.

Hong, E. R., Gong, L. Y., Ninci, J., Morin, K., Davis, J. L., Kawaminami, S., … & Noro, F. (2017). A meta-analysis of single-case research on the use of tablet-mediated interventions for persons with ASD. *Research in Developmental Disabilities, 70,* 198–214. https://doi.org/10.1016/j.ridd.2017.09.013.

Horner, R. H., Carr, E. G., Halle, J., McGee, G., Odom, S., & Wolery, M. (2005). The use of single-subject research to identify evidence-based practice in special education. *Exceptional Children, 71*, 165–179. https://doi.org/10.1177/001440290507100203.

Hughes, C. (2001). Transition to adulthood: supporting young adults to access social, employment, and civic pursuits. *Mental Retardation and Developmental Disabilities Research Reviews, 7*, 84–90. https://doi.org/10.1002/mrdd.1012.

*Hume, K., & Odom, S. (2007). Effects or an individual work system on the independent functioning of students with autism. Journal of Autism and Developmental Disorders, 37, 1166–1180. https://doi.org/10.1007/s10803-006-0260-5.

Individuals with Disabilities Education Improvement Act (IDEA) of 2004, PL 108-446, 20 U.S.C. ss 1,400 et seq.

Johnson, D. R. (2012). Policy and adolescent transition education. In M. L. Wehmeyer & K. W. Webb (Eds.), *Handbook of adolescent transition education for youth with disabilities* (pp. 11–31). New York, NY: Routledge.

Kagohara, D. M., Van der Meer, L., Ramdoss, S., O'Reilly, M. F., Lancioni, G. E., Davis, T. N., … Sigafoos, J. (2013). Using iPods ® and iPads ® in teaching programs for individuals with developmental disabilities: a systematic review. *Research in Developmental Disabilities, 34*, 147–156. https://doi.org/10.1016/j.ridd.2012.07.027.

Kazdin, A. E. (2011). *Single-case research designs: methods for clinical and applied settings* (2nd ed.). New York, NY: Oxford University Press.

*Kelly, J.A., Wildman, B.G., & Berler, E.S. (1980). Small group behavioral training to improve the job interview skills repertoire of mildly retarded adolescents. Journal of Applied Behavior Analysis, 13, 461–471. https://doi.org/10.1901/jaba.1980.13-461.

*Kemp, D.C., & Carr, E.G. (1995). Reduction of severe problem behavior in community employment using an hypothesis-driven multicomponent intervention approach. Journal of the Association of Persons with Severe Handicaps, 20, 229–247. https://doi.org/10.1177/154079699602000401.

Kratochwill, T. R., Hitchcock, J., Horner, R. H., Levin, J. R., Odom, S. L., Rindskopf, D. M., & Shadish, W. R. (2010). Single-case design technical documentation. Retrieved August 1, 2015, from What Works Clearinghouse website: http://ies.ed.gov/ncee/wwc/pdf/wwc_scd.pdf.

Kratochwill, T. R., Hitchcock, J. H., Horner, R. H., Levin, J. R., Odom, S. L., Rindskopf, D. M., & Shadish, W. R. (2013). Single-case intervention research design standards. *Remedial and Special Education, 34*, 26–38. https://doi.org/10.1177/0741932512452794.

Kruskal, W. H., & Wallis, A. W. (1952). Use of ranks in one-criterion variance analysis. *Journal of the American Statistical Association, 47*, 583–621. https://doi.org/10.2307/2280779.

*Lattimore, L.P., Parsons, M.B., & Reid, D. (2008). Simulation training of community job skills for adults with autism: a further analysis. Behavior Analysis in Practice, 1, 24–29. https://doi.org/10.1007/bf03391717.

*Lattimore, L.P., Parsons, M.B., & Reid, D. (2009). Rapid training of community job skill to nonvocal adults with autism: an extension of intensive training. Behavior Analysis in Practice, 2, 34–42. https://doi.org/10.1007/bf03391735.

*Lattimore, L.P., Parsons, M.B., & Reid, D.H. (2006). Enhancing job-site training of supported workers with autism: a reemphasis on simulation. Journal of Applied Behavior Analysis, 39, 91–102. https://doi.org/10.1901/jaba.2006.154-04.

*Likins, M., Salzberg, C.L., Stowitschek, J.J., Lignugaris-Kraft, & Curl, R. (1989). Co-worker implemented job training: the use of coincidental training and quality-control checking on the food preparation skills of trainees with mental retardation. Journal of Applied Behavioral Analysis, 22, 381–393. https://doi.org/10.1901/jaba.1989.22-381.

Lipsey, M. W., & Wilson, D. B. (2001). Applied social research methods series. In L. Bickman & D. J. Rog (Eds.), *Practical meta-analysis* (Vol. 49). Thousand Oaks, CA: Sage.

Maggin, D. M., & Odom, S. L. (2014). Evaluating single-case research data for systematic review: a commentary for the special issue. *Journal of School Psychology, 52*, 237–241. https://doi.org/10.1016/j.jsp.2014.01.002.

Maggin, D. M., Briesch, A. M., & Chafouleas, S. M. (2013). An application of the What Works Clearinghouse Standards for evaluating single-subject synthesis of the self-management literature base. *Remedial and Special Education, 34*, 44–58. https://doi.org/10.1177/0741932511435176.

*Martin, J.E., Elias-Burger, S., & Mithuag, D.E. (1987). Acquisition and maintenance of time-based task change sequence. Education and Training in Mental Retardation, 22, 250–255.

Mason, R. A., Ganz, J. B., Parker, R. I., Burke, M. D., & Camargo, S. P. (2012). Moderating factors of video-modeling with other as model: a meta-analysis of single-case studies. *Research in Developmental Disabilities, 33*, 1076–1086. https://doi.org/10.1016/j.ridd.2012.01.016.

Mason, R. A., Davis, H. S., Boles, M. B., & Goodwyn, F. (2013). Efficacy of point-of-view video modeling: a meta-analysis. *Remedial and Special Education, 34*, 333–345. https://doi.org/10.1177/0741932513486298.

*Mechling, L.C., & Ayres, K.M. (2012). A comparative study: completion of fine motor office related tasks by high school students with autism using video models on large and small screen sizes. Journal of Autism and Developmental Disorders, 42, 2364–2373. https://doi.org/10.1007/s10803-012-1484-1.

*Mechling, L.C., & Ortega-Hurndon, F. (2007). Computer-based video instruction to teach young adults with moderate intellectual disabilities to perform multiple step, job tasks in a generalized setting. Education and Training in Developmental Disabilities, 42, 24–37.

*Mechling, L.C., & Savidge, E.J. (2011). Using a personal digital assistant to increase completion of novel tasks and independent transitioning by students with autism spectrum disorder. Journal of Autism and Developmental Disorders, 41, 687–704. https://doi.org/10.1007/s10803-010-1088-6.

Migliore, A., & Butterworth, J. (2008). Trends in outcomes of the vocational rehabilitation program for adults with developmental disabilities: 1995-2005. *Rehabilitation Counseling Bulletin, 52*, 35–44. https://doi.org/10.1177/0034355208320075.

*Mitchell, R.J., Schuster, J.W., Collins, B.C., & Gassaway, L.J. (2000). Teaching vocational skills with a faded auditory prompting system. Education and Training in Mental Retardation and Developmental Disabilities, 35, 415–427.

*Morgan, R.L., & Salzberg, C.L. (1992). Effects of video-assisted training on employment-related social skills of adults with severe mental retardation. Journal of Applied Behavior Analysis, 25, 365–383. https://doi.org/10.1901/jaba.1992.25-365.

No Child Left Behind (NCLB) Act of 2001, Pub. L. No. 107-110, sec. 115, Stat. 1425 (2002).

Newman, L., Wagner, M., Cameto, R., & Knokey, A.-M. (2009). The post-high school outcomes of youth with disabilities up to 4 years after high school. A report of findings from the National Longitudinal Transition Study-2 (NLTS2) (NCSER 2009–3017). Menlo Park, CA: SRI International. Retrieved August 1, 2015, from www.nlts2.org/reports/2009_04/nlts2_report_2009_04_complete.pdf.

Newman, L., Wagner, M., Knokey, A.-M., Marder, C., Nagle, K., Shaver, D., … Schwarting, M. (2011). *The post-high school outcomes of young adults with disabilities up to 8 years after high school: a report from the National Longitudinal Transition Study-2 (NLTS2)* (NCSER 2011–3005). Menlo Park, CA: SRI International. Retrieved August 1, 2015, from www.nlts2.org/reports/ .

Ninci, J., Neely, L. C., Hong, E. R., Boles, M. B., Gilliland, W. D., Ganz, J. B., … Vannest, K. J. (2015). Meta-analysis of single-case research on teaching functional living skills to individuals with ASD. *Review Journal of Autism and Developmental Disorder, 2*, 184–198. https://doi.org/10.1007/s40489-014-0046-1.

Palmen, A., Didden, R., & Lang, R. (2012). A systematic review of behavioral intervention research in adaptive skill building in high functioning young adults with autism spectrum disorder. *Research in Autism Spectrum Disorders, 6*, 602–617. https://doi.org/10.1016/j.rasd.2011.10.001.

Palmer, R. (2014). Technical and vocational skills and post-2015: avoiding another vague skills goal? *International Journal of Educational Development, 39*, 32–39. https://doi.org/10.1016/j.ijedudev.2014.08.007.

Parker, R. I., & Vannest, K. J. (2009). An improved effect size for single-case research: nonoverlap of all pairs. *Behavior Therapy, 40*, 357–367. https://doi.org/10.1016/j.beth.2008.10.006.

Parker, R. I., Vannest, K. J., & Davis, J. L. (2011a). Effect size in single-case research: a review of nine nonoverlap techniques. *Behavior Modification, 35*, 303–322. https://doi.org/10.1177/0145445511399147.

Parker, R. I., Vannest, K. J., Davis, J. L., & Sauber, S. B. (2011b). Combining nonoverlap and trend for single-case research: Tau-U. *Behavior Therapy, 42*, 284–299. https://doi.org/10.1016/j.beth.2010.08.006.

*Parson, M.B., Reid, D.H., Green, C.W., Browning, L.B. (1999). Reducing individualized job coach assistance provided to persons with multiple severe disabilities in supported work. Research and Practice for Persons with Severe Disabilities, 24, 292–297. https://doi.org/10.2511/rpsd.24.4.292.

*Riffel, L.A., Wehmeyer, M.L., Turnbull, A.P., Lattimore, J., Davies, D., Stock, S., & Fisher, S. (2005). Promoting independent performance of transition-related tasks using a palmtop PC-based self-directed visual and auditory prompting system. Journal of Special Education Technology, 20, 5–14. https://doi.org/10.1177/016264340502000201.

Roth, M. E., Gillis, J. M., & Reed, F. D. D. (2014). A meta-analysis of behavioral interventions for adolescents and adults with autism spectrum disorders. *Journal of Behavioral Education, 23*, 258–286. https://doi.org/10.1007/s10864-013-9189-x.

Rusch, F. R., & Dattilo, J. (2012). Employment and self-management: a meta-evaluation of seven literature reviews. *Intellectual and Developmental Disabilities, 50*, 69–75. https://doi.org/10.1352/1934-9556-50.1.69.

Sanford, C., Newman, L., Wagner, M., Cameto, R., Knokey, A.-M., & Shaver, D. (2011). *The post-high school outcomes of young adults with disabilities up to 6 years after high school. Key findings from the National Longitudinal Transition Study-2 (NLTS2)* (NCSER 2011–3004). Menlo Park, CA: SRI International. Retrieved August 1, 2015, from www.nlts2.org/reports.

Soares, D. (2011). Effect size and moderators of effects for token economy interventions. Doctoral dissertation, Texas A&M University. Retrieved August 1, 2015, from http://hdl.handle.net/1969.1/ETD-TAMU-2011-12-10436 .

Taylor, J. L., McPheeters, M. L., Sathe, N. A., Dove, D., Veenstra-VanderWeele, J., & Warren, Z. (2012). A systematic review of vocational interventions for young adults with autism spectrum disorders. *Pediatrics, 130*, 531–538. https://doi.org/10.1542/peds.2012-0682.

Test, D. W., Richter, S., & Walker, A. R. (2012). Life skills and community-based instruction. In M. L. Wehmeyer & K. W. Webb (Eds.), *Handbook of adolescent transition education for youth with disabilities* (pp. 121–138). New York, NY: Routledge.

*Van Laarhoven, T., Van Laarhoven-Myers, T., & Zurita, L.M. (2007). The effectiveness of using a pocket PC as a video modeling and feedback device for individuals with developmental disabilities in vocational settings. Assistive Technology Outcomes and Benefits, 4, 28–45.

*Wacker, D.P., Berg, W.K., Berrie, P., & Swatta, P. (1985). Generalization and maintenance of complex skills by severely handicapped adolescents following picture prompt training. Journal of Applied Behavior Analysis, 18, 329–226. https://doi.org/10.1901/jaba.1985.18-329.

*Wacker, D.P., Berg, W.K., Choisser, L., & Smith, J. (1989). Evaluation of the generalize effects of a peer-training procedure with moderately retarded adolescents. Journal of Applied Behavior Analysis, 22, 261–273. https://doi.org/10.1901/jaba.1989.22-261.

Walker, A. R., Richter, S., Uphold, N. M., & Test, D. W. (2010). Review of the literature on community-based instruction across grade-levels. *Education and Training in Autism and Developmental Disabilities, 45*, 242–267.

Wehmeyer, M. L., Palmer, S. B., Smith, S. J., Parent, W., Davies, D. K., & Stock, S. (2006). Technology use by people with intellectual and developmental disabilities to support employment activities: a single-subject design meta analysis. *Journal of Vocational Rehabilitation, 24*, 81–86.

Whalon, K. J., Conroy, M. A., Martinez, J. R., & Werch, B. L. (2015). School-based peer-related social competence interventions for children with autism spectrum disorder: a meta-analysis and descriptive review of single case research design studies. *Journal of Autism and Developmental Disorders, 45*, 1513–1531. https://doi.org/10.1007/s10803-015-2373-1.

What Work Requires of Schools: A SCANS Report for America 2000 (1991). The Secretary's Commission on achieving necessary skills. US Department of Labor

Wilcoxon, F., Rhodes, L. J., & Bradley, R. A. (1963). Two sequential two-sample grouped rank tests with applications to screening experiments. *Biometrics, 19*, 58–77. https://doi.org/10.2307/2527572.